AIT · NATURAL LANGUAGE PROCESSING · GROUP 12

EpiCite

A Two-Stage Epistemic Sentence Classifier and Citation-Need Scorer with Triangulated Feature Attribution

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Why citation-need prediction needs a richer signal

The state of the art is binary

Redi et al. (2019) frame citation-need as binary classification over surface features (length, position). It works on Wikipedia, but cannot tell us why a sentence was flagged.

What gets lost

An unverified factual Claim, a subjective Opinion, and an encyclopedic Background statement all look the same to such a model — yet they need very different treatment.

Our angle

Predict the epistemic role first, then score citation need — and instrument every step with attribution methods so a reviewer can audit each flag.

Four epistemic roles

CLAIM

An assertion of fact requiring support.

EVIDENCE

Cites data, studies, or named sources.

OPINION

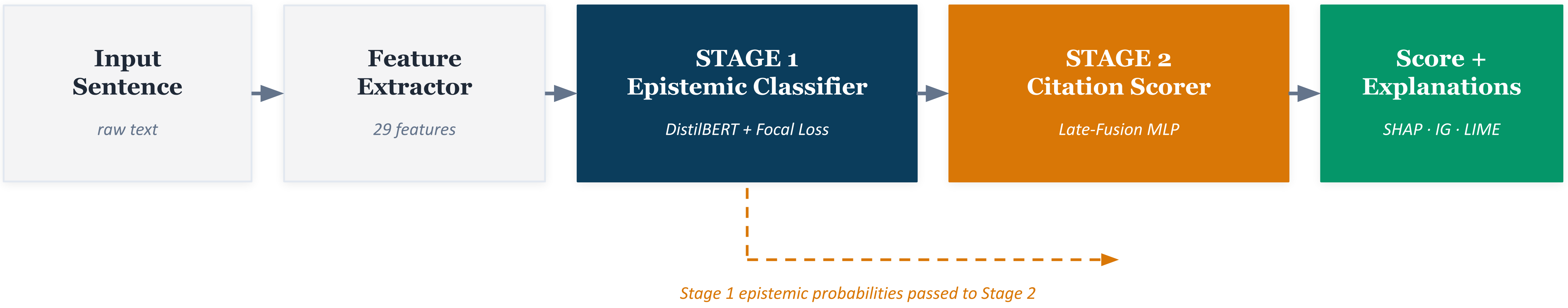
Subjective stance, not citable as fact.

BACKGROUND

Common knowledge or framing.

The two-stage EpiCite pipeline

A sentence flows from input to a citation-need score, plus an epistemic label, plus per-token and per-feature attributions.



Class-imbalanced training

Focal loss (γ=2) + WeightedRandomSampler beat 65× Claim/Evidence imbalance.

Late-Fusion design

[CLS] (768-d) ⊕ 29 engineered features ⊕ 5 epistemic probs → 3-layer MLP.

Triangulated attribution

SHAP + Permutation + MI aggregated by Borda rank, plus token-level IG / LIME.

Data: combining IBM, UKP, and WikiSQE

Stage 1 — Epistemic labels

IBM Claim Detection + IBM Evidence + IBM CDC + UKP Argument Mining → 4-class taxonomy.

Class	Original	+ IBM	Δ
Claim	13,225 (48.3%)	17,388 (49.2%)	+4,163
Evidence	204 (0.7%)	3,949 (11.2%)	+3,745
Opinion	2,433 (8.9%)	2,433 (6.9%)	0
Background	11,536 (42.1%)	11,536 (32.7%)	0
Total	**27,398**	**35,306**	**+7,908**

65× imbalance, mitigated by IBM Evidence (+3,745) and class-balanced sampling.

Stage 2 — Citation labels

WikiSQE — citation-needed config, official splits.

151,981 WikiSQE sentences (all splits)

150,000 balanced training (75k pos / 75k neg)

991 official test set for headline metrics

Stage 1 — Focal-loss epistemic classifier

0.763

Macro-F1

combined test set

78.7%

Accuracy

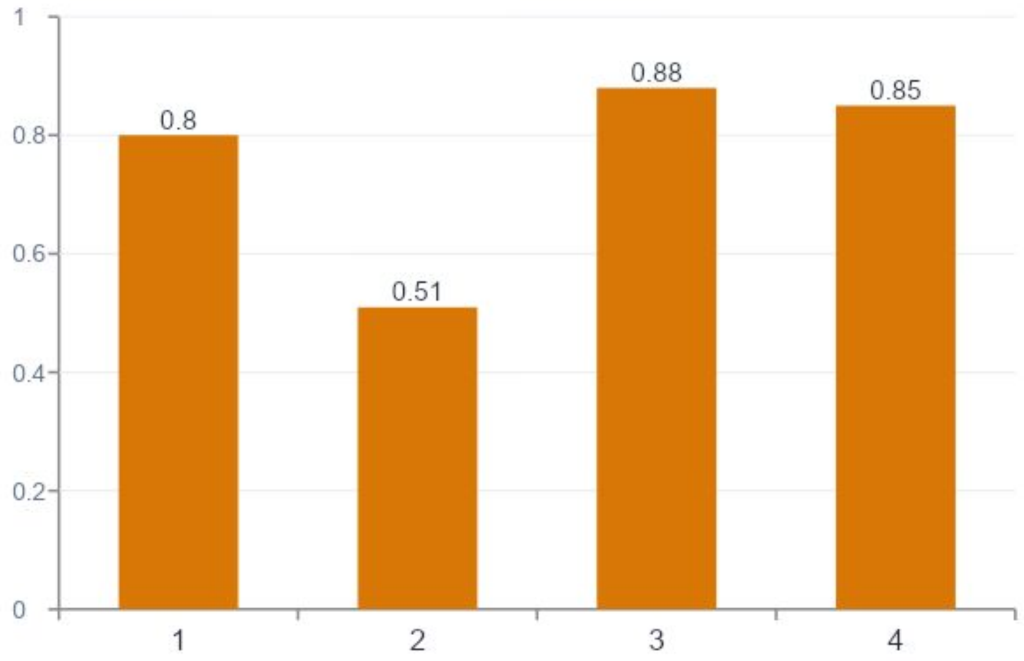
n = 3,531

+0.27

Evidence F1 lift

vs unweighted baseline

Per-class F1 score



Focal loss does the heavy lifting

γ=2 + α-weights down-weight easy Claims so the rare Evidence gradient dominates training.

IBM augmentation rescues Evidence

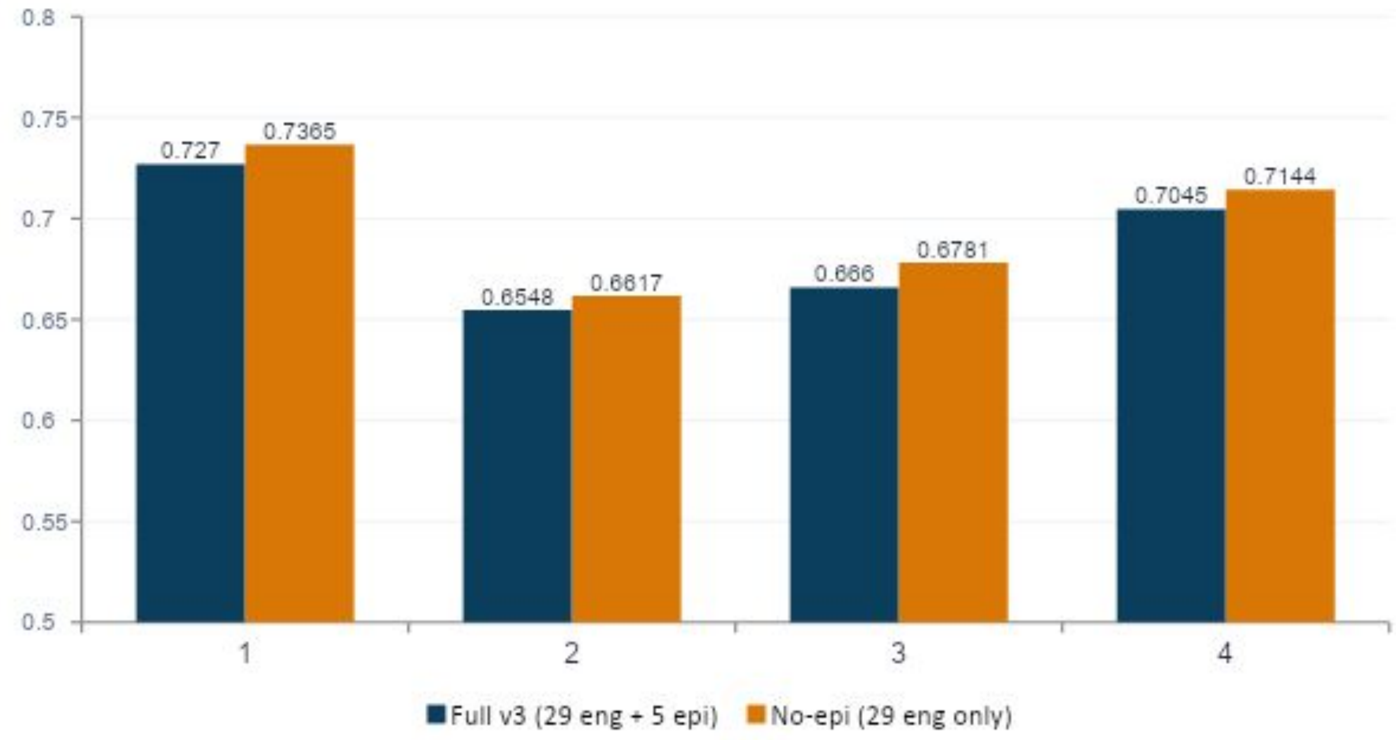
Evidence sample count grew from 204 to 3,949; F1 jumped from collapse to 0.51.

OOD calibration is the soft spot

On the IBM-only held-out set the absolute probabilities don't transfer — relative ranking does.

Stage 2 — and the surprise ablation

Late-Fusion variants on the WikiSQE test split (n = 991)



THE FINDING

Removing the Stage 1 signal improves the scorer.

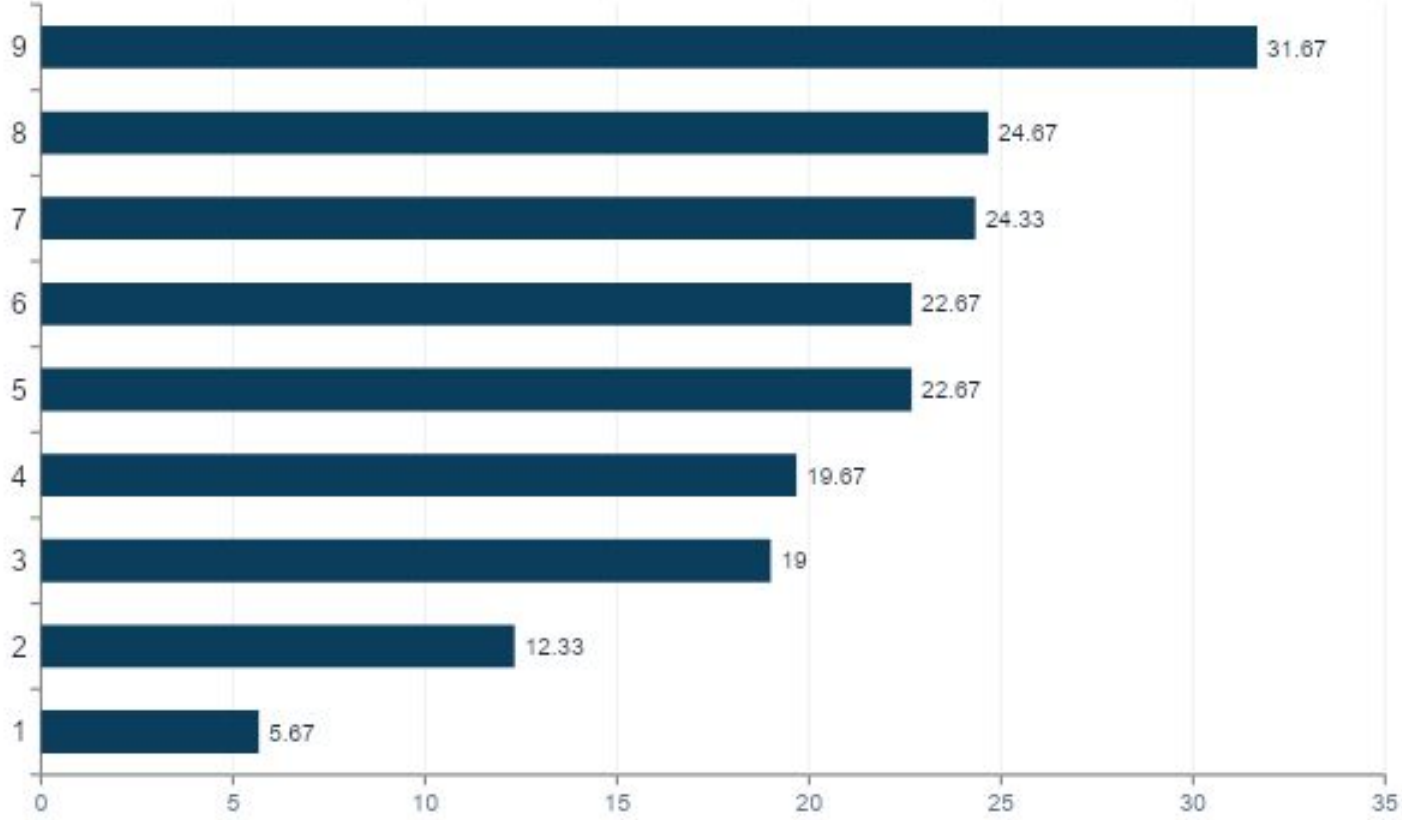
– 0.0095 AUC

when adding the 5-dim epistemic probability vector to the Stage 2 input.

Likely cause: the [CLS] embedding already encodes most of the epistemic signal — the explicit handoff just adds noise.

Which features actually matter? Triangulated.

Three independent methods (SHAP, permutation, mutual information) ⟶ Borda-rank aggregation. Lower rank = more consistently important.



H1 — REJECTED

Epistemic features do NOT outrank surface features.

epi_prob_Claim ranks #30 of 34. epi_prob_Evidence ranks #33. sentence_length is unanimously #1.

BUT — class-conditionally

Different epistemic classes are driven by different features:

Claim ⟶ n_numbers, ent_total
Evidence ⟶ length, hedge_verbs
Background ⟶ length, ent_date

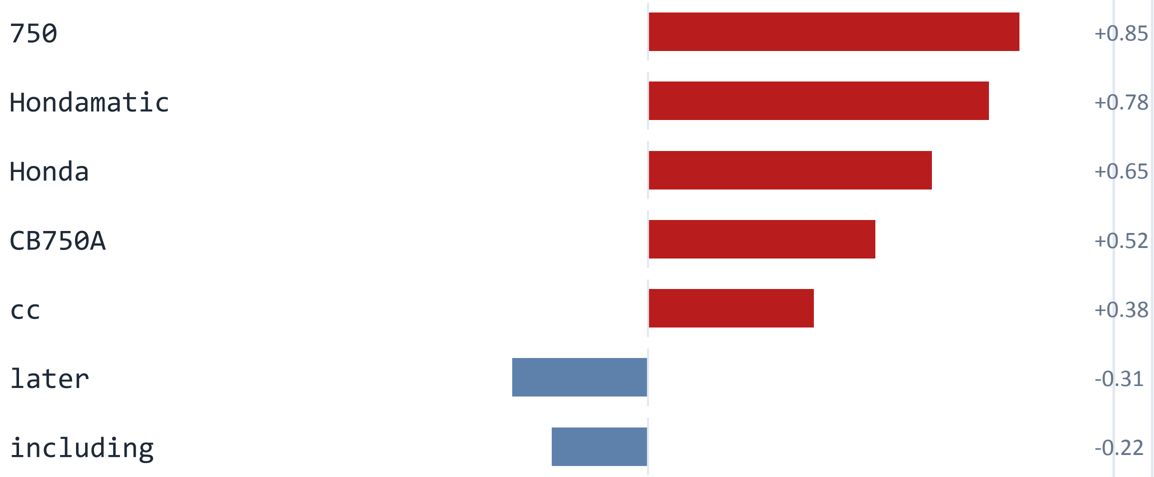
Inside one prediction — IG + LIME

HIGH-CONFIDENCE FALSE POSITIVE · predicted score 0.977 · true label = no citation

“The Hondamatic was later used in Honda’s 400, 450 and 750 cc motorcycles, including the CB750A.”

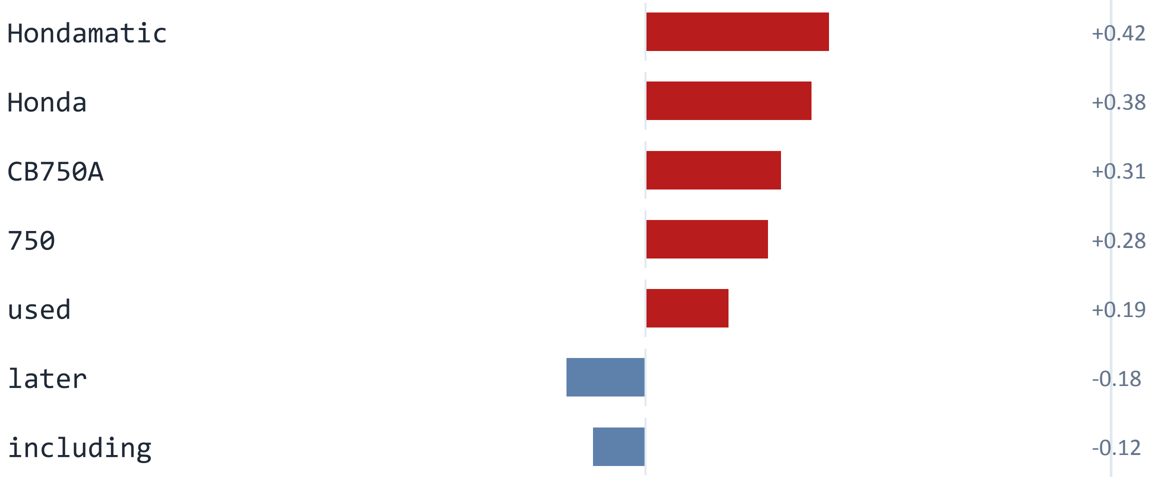
INTEGRATED GRADIENTS

token-level · gradient-based



LIME

token-level · perturbation-based (n_samples = 300)



■ pushes score UP (toward “citation needed”) ■ pushes score DOWN *Both methods agree: numbers and named entities triggered the wrong flag.*

When EpiCite over-flags: false positives

High-confidence false positives from the WikiSQE test split, with the engineered features that drove each score up (per-sentence SHAP).

0.996 FALSE POSITIVE

“Flag of Turkey December 23 — An Ankara-Samsun regular route bus head-on collided with a fuel truck and burning both vehicle at outskirt of …”

TOP FEATURES PUSHING SCORE UP

`rate_ent_total`	=0.34	+0.21
`n_proper_nouns`	=5	+0.16
`sentence_length`	=29	+0.14

News-style headline fragment, dense in named entities and numbers — the model can’t tell it from a citable claim.

0.977 FALSE POSITIVE

“The Hondamatic was later used in Honda’s 400, 450 and 750 cc motorcycles, including the CB750A.”

TOP FEATURES PUSHING SCORE UP

`n_numbers`	=5	+0.18
`rate_ent_total`	=0.21	+0.10
`sentence_length`	=19	+0.12

Encyclopedic factual recall — already considered common knowledge in the article context, but looks numeric and entity-rich.

0.987 FALSE POSITIVE

“She is perhaps best known for her second-wave feminism work, though she says, ‘My politics are much broader than that.’…”

TOP FEATURES PUSHING SCORE UP

`n_hedge_adv`	=2	+0.13
`n_subjective_pron`	=3	+0.11
`sentence_length`	=27	+0.13

Mixed opinion + biographical claim. Hedging (“perhaps”) + length push the score up, even though the claim is sourced via direct quote.

PATTERN: *long, entity-rich, number-dense sentences are flagged regardless of whether they’re common knowledge or hedged — directly mirroring the global SHAP top features.*

When EpiCite misses: false negatives

High-confidence false negatives — sentences EpiCite scored near-zero, where WikiSQE flagged them as needing a citation.

0.000 FALSE NEGATIVE

"Michael Barker, Regulating revolutions in Eastern Europe: Polyarchy and the National Endowment for Democracy, 1 November 2006."

PATTERN

Bibliographic reference

Reads like a reference list entry, not a sentence. Lacks a verb, lacks the discursive structure the model learned to associate with citable Claims.

0.001 FALSE NEGATIVE

"Betty Krawczyk ran unsuccessfully in the riding of Vancouver East, receiving 1.02% of the votes (423 votes)."

PATTERN

Election result fragment

A clear factual statistic that should be sourced. The model treats it as encyclopedic Background — short, entity + number, but no surface markers of contention.

0.002 FALSE NEGATIVE

"He was drafted in the second round (50th overall) of the 2005 Major League Baseball Draft by the Kansas City Royals."

PATTERN

Sports stat

Specific draft position with no hedging or causal markers. The model has learned that bare biographical facts of this shape don't need flags.

CASCADE: *93.7% of these misses had high Stage 1 confidence — both stages share the same blind spot for bibliographic and list-fragment text.*

Head-to-head vs Qwen-0.5B-Instruct

ROC-AUC GAIN

+0.19

EpiCite 0.727 vs Qwen 0.539

PARAM RATIO

3.7×

Qwen has 3.7× more params and still loses

LATENCY ADVANTAGE

11×

0.78 ms vs 8.43 ms per sentence

Three-axis comparison

Model	Params	AUC	F1	Latency	Interpretability
EpiCite (ours)	66.5 M	**0.727**	**0.655**	0.78 ms	✓ SHAP · IG · LIME · class-cond. · calibration
Qwen-0.5B-Instruct	494 M	0.539	0.239	8.43 ms	✗ Black-box; CoT is post-hoc, not faithful

Cohen's κ on error sets = −0.005 → near-independent failures, suggesting ensemble potential.

Hypotheses: two rejections, one supported

H1

REJECTED

Epistemic type outranks surface features (e.g. length).

sentence_length is the unanimous #1 feature across SHAP, permutation, and MI. epi_prob_Claim ranks #30 of 34.

H2

SUPPORTED

Two-stage Late-Fusion offers higher interpretability than end-to-end.

We deliver three feature-level views (SHAP / Perm / MI), one token-level (IG), one class-conditional, plus the audit and MCP endpoints.

H3

REJECTED

Removing the Stage 1 signal degrades scorer performance.

Removing it slightly improves every metric (−0.010 AUC). Information is redundant with the shared [CLS] encoder.

From research artefact to working tool

USE CASE 1

Wikipedia article audit

✓	–	The Eiffel Tower was completed in 1889 …
✗	FLAGGED	Many visitors find it beautiful at night.
✓	FLAGGED	Studies show ~7 million people visit each year.
✗	FLAGGED	It is widely considered an icon of France.
✓	FLAGGED	Some experts argue its design influenced 20th-c. architecture.

USE CASE 2

MCP server walkthrough

```
analyze_sentence(s)
→ epi_class: "Evidence"
→ citation_score: 0.845
→ needs_citation: true

analyze_paragraph(p)
→ n_flagged: 2 of 4
→ mean_score: 0.572

explain_prediction(s)
→ method: integrated_gradients
→ top tokens with weights
```

What we'd do next, honestly

01 Cross-domain transfer

All numbers live on Wikipedia. News and student-essay sets need their own annotation pass and per-domain calibration.

02 Calibration via temperature scaling

Both variants are over-confident (ECE $\geq$ 0.27). A single scalar fit on val should fix this without retraining.

03 Annotation noise estimate

A 40-sentence pilot of borderline disagreements suggests non-trivial WikiSQE label noise. Hand-coding remains.

04 Ensembling with Qwen

Cohen's $\kappa = -0.005$ on errors → an ensemble could plausibly recover ~70% combined coverage at modest extra cost.

THREE TAKEAWAYS

1

Triangulation matters.

A single attribution method would have credited length alone or boosters alone. Three methods + Borda gave us a defensible ranking.

2

Negative ablations are publishable.

The Stage-1 handoff is redundant with [CLS] — we report this honestly and reposition EpiCite as interpretability-first.

3

Small + interpretable beats big + opaque (here).

+0.19 AUC over Qwen-0.5B at 11× lower latency, with full SHAP / IG / LIME instrumentation.

Thank you — questions welcome.

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